

PrecipQC

Gauge Reading Image Quality Checker

User Manual

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This manual explains how to prepare the input data and use the interface to verify cup rain gauge readings against photographic records.

Browser required: Google Chrome or Microsoft Edge

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1. Overview

The PrecipQC tool is a web-based application that helps you check whether the cup rain gauge readings recorded in your data match the photographs taken at each station. It runs entirely in your web browser - no internet connection is required once the file is open, and nothing is uploaded to any server.

The tool works by showing you, for each station and date, three things side by side:

- The value recorded in your CSV file (cup gauge reading)
- The tipping bucket reading for the same date, if available
- The photograph(s) of the gauge taken on that date

For each record, you look at the photograph, compare it to the recorded value, and press one of three buttons: MATCH, UNSURE, or WRONG. At the end of the session, the tool produces output files containing only the verified data.

2. Preparing the Input Files

You need to prepare two CSV files before opening the tool.

Tip: To save a file as CSV from Excel: go to File → Save As → choose the folder → in the “Save as type” dropdown, select CSV (Comma delimited) → click Save.

2.1 Cup gauge CSV (required)

This file contains your cup rain gauge readings and the filenames of the associated photographs. Every row represents one reading at one station on one date.

File preparation is most easily done by renaming columns, formatting numbers and removing redundant columns in Excel. The file must have exactly these five columns, with these exact names in the first row:

Column name	Example	Notes
date	2026-02-20	The date of the reading. Change the format if necessary.
location	Garizegu	Name of the station. Spelling must be consistent across all rows. Capital letters do not matter — the tool will match them automatically. Rename ‘uds_wacwisa’ to ‘Kyankpala’.
gauge_mm	14.5	The cup gauge reading in mm. Use a decimal point, not a comma (e.g. 14.5 not 14,5). For zero rain, enter 0.
image1	1771572962720.jpg	The filename of the first image, including .jpg at the end. Do not include any folder path, just the filename.

image2	1771571902373.jpg	The filename of a second image, if a second photo was taken for the same reading. Leave this cell blank if there is only one image.
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Note: Column names are **case-sensitive**. Write them exactly as shown above: all lowercase, with underscores, not spaces. The first row of the file must be the header row — do not leave any blank rows at the top.

	A	B	C	D	E	F
1	date	location	gauge_mm	image1	image2	
2	2/20/26	Garizegu	0	1771572962720.jpg		
3	2/20/26	Gbullung	0	1771571879.jpg	1771571902373.jpg	
4	2/20/26	Sanga	0	1771570846639.jpg		
5	2/20/26	Tuunaayili	0	1771569753055.jpg		
6	2/19/26	Tingoli	0	1771495798341.jpg		
7	2/19/26	Sanga	0	1771485626588.jpg		
8	2/19/26	Garizegu	0	1771485579101.jpg		
9	2/19/26	Gbullung	0	1771485223.jpg	1771485242692.jpg	
10	2/19/26	Tuunaayili	0	1771482602037.jpg		
11	2/18/26	Tingoli	0	1771414356891.jpg		
12	2/18/26	Garizegu	0	1771398774847.jpg		
13	2/18/26	Sanga	0	1771398242665.jpg		
14	2/18/26	Gbullung	0	1771398199.jpg	1771398215984.jpg	
15	2/18/26	Tuunaayili	0	1771396329680.jpg		
16	2/17/26	Tingoli	0	1771317337684.jpg		
17	2/17/26	Kpaliga	0	1771313394889.jpg		
18	2/17/26	Gbullung	0	1771311729.jpg	1771311748092.jpg	

Figure 1. Example of a cup gauge input CSV file

2.2 Bucket CSV (optional)

This file contains the tipping bucket logger data. It is optional. If you do load it, the tool will display the corresponding tipping bucket reading alongside the cup value for each record, which can help you identify suspicious entries.

Important Note About Date Cutoff for Tipping Bucket Data

The tipping bucket logger records timestamps continuously throughout the day. To match the **readings from cup gauges** (recorded and emptied at around 7 AM every morning), all bucket events are automatically shifted by **17 hours** in the processing step.

This means:

- Any tip recorded **before 7:00 AM** is assigned to the **current calendar day**
- Any tip recorded **at or after 7:00 AM** is assigned to the **next calendar day**

This ensures the tipping bucket daily totals align perfectly with the morning reading schedule used for the cup gauge, so both datasets use the same date definition.

File preparation is most easily done using the Python script (please refer to [Annex A](#)). The file must have exactly these four columns:

Column name	Example	Notes
location	GARIZEGU	Name of the station in capital letters. The tool will match this to the gauge data regardless of capitalisation, so GARIZEGU will match Garizegu.
date	2026-02-20	The date of the reading. Must follow the same format as the gauge CSV.
tips	36	The number of tips recorded by the tipping bucket logger on that day.
bucket_mm	7.2	The total rainfall in millimetres calculated from the tips. Number of tips × 0.2 mm per tip.

Note: Days where there is only tipping bucket data and no cup reading will be automatically excluded from the review. Only records with at least one photograph are included.

	A	B	C	D
1	location	date	tips	bucket_mm
2	GALINKPEGU	2/24/25	1	0.2
3	GALINKPEGU	3/6/25	2	0.4
4	GALINKPEGU	3/13/25	8	1.6
5	GALINKPEGU	3/15/25	36	7.2
6	GALINKPEGU	3/19/25	1	0.2
7	GALINKPEGU	4/10/25	3	0.6
8	GALINKPEGU	4/11/25	47	9.4
9	GALINKPEGU	4/17/25	18	3.6
10	GALINKPEGU	4/19/25	1	0.2
11	GALINKPEGU	4/23/25	2	0.4
12	GALINKPEGU	4/27/25	1	0.2

Figure 2. Example of a tipping bucket input CSV file

2.3 Image Folder

All your photographs must be stored in a single folder on your computer. The filenames in the images must match exactly what is written in the image1 and image2 columns of the gauge CSV, including the .jpg extension.

2.4 Common Formatting Mistakes to Avoid

- Using a comma as a decimal separator in gauge_mm (e.g. 14,5 instead of 14.5)
- Extra spaces before or after values in cells
- Blank rows in the middle of the data

3. Opening the Tool

The tool is a single file named PrecipQC.html. You do not need to install any software to use it.

1. Locate the file PrecipQC_v4.html on your computer.
2. Right-click the file and choose **Open with** → **Google Chrome** or **Microsoft Edge**. Do not open it with Firefox, Safari, or Internet Explorer.
3. The Precip QC setup screen will appear in your browser window.

Note: If double-clicking the file opens it in the wrong browser, right-click and use “Open with” to choose the correct one.

4. Loading Your Files (Setup Screen)

When the tool opens, you will see a setup screen with three steps.

Step 1 — Load the Gauge CSV

1. Click the **Browse** button next to **Cup gauge CSV**.
2. A file picker window will open. Navigate to your gauge CSV file and click Open.
3. The filename will appear below the button, and a green tick (✓) will appear on the right to confirm it has been loaded successfully.

Step 2 — Load the Bucket CSV (optional)

4. Click **Browse** next to **Bucket CSV** if you have tipping bucket data.
5. Select your bucket CSV file and click Open.
6. If you do not have bucket data, you can skip this step entirely. The tool will still work — the “Tipping bucket” field will show **No data** for all records.

Step 3 — Select the Image Folder

7. Click **Browse** next to **Image Folder**.
8. A folder picker window will open. Navigate to the folder containing your photographs and click Select Folder.
9. Your browser may ask for permission to access the folder — click Allow or Yes.
10. The folder name will appear below the button with a green tick.

Step 4 — Select dates to review

11. If desired, select start and end dates for what you want to review in this session.

Starting the Review

Once the Cup gauge CSV and Image Folder are both loaded (the Bucket CSV is optional), the **Start Review** → button at the bottom will become active. Click it to begin.

5. The Review Screen

After clicking Start Review, the main review screen opens. It is divided into two parts: a panel on the left showing the data for the current record, and a large image area on the right showing the photograph(s).

5.1 The Top Bar

The top bar runs across the full width of the screen and shows three things at all times:

- **Progress bar and counter** — shows how many records you have reviewed out of the total (e.g. 47 / 350). The blue bar fills up as you confirm decisions.
- **Filter button** — opens a panel to select specific stations to review. Only records from your chosen stations will appear in the workflow, while preserving all existing decisions.
- **Jump to button** — lets you quickly navigate to any record by entering a full or partial date (YYYY-MM-DD), skipping directly to the first matching entry.
- **Colour tallies** — three small badges showing the running count of MATCH (green), UNSURE (yellow), and WRONG (red) decisions made so far.
- **End & Export button** — in the top-right corner. Click this at any time to finish the session and download your output files.

5.2 The Left Panel — Record Information

The left panel shows the data for the record currently on screen.

Record section

- **Station** — the name of the monitoring station.
- **Date** — the date of the reading.

Readings section

- **Cup gauge (mm)** — the value recorded in the gauge CSV for this station and date. This is the number you are checking against the photograph.
- **SPLIT READING badge** — if this appears, the tool has detected multiple rows for the same station and date. For example, when the gauge overflowed and was emptied before a second reading was taken for the remaining water. The individual sub-readings (e.g. Row 1: 26.0 mm, Row 2: 14.0 mm) are shown below the badge. The highest value is displayed by default.



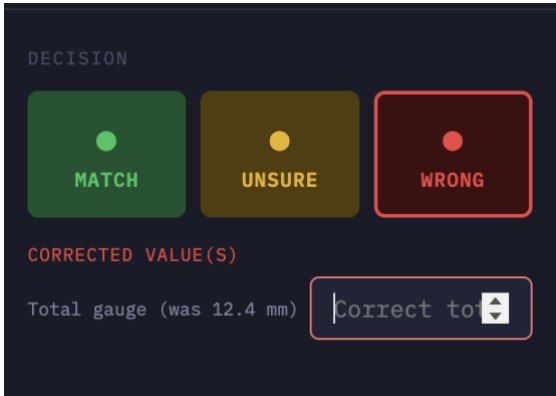
- **Tipping bucket (mm)** — the corresponding tipping bucket reading, if available. Shows “No data” if no bucket data was loaded or if there is no record for this date.
- **Discrepancy** — the difference between the cup and bucket readings (cup minus bucket). Shown in green if small, yellow if moderate, and red if large. A large discrepancy does not necessarily mean the reading is wrong, but it is a useful flag to look more carefully at the image.

Already-reviewed banner

If you navigate back to a record you have already confirmed, a coloured banner will appear showing the decision you recorded (e.g. “✓ Confirmed: Match”). You can change the decision by clicking a different button and confirming again.

5.3 Making a Decision

After examining the photograph, press one of the three decision buttons:

Button	Meaning	When to use and action needed
MATCH (number key 1)	The image clearly shows the cup gauge, and the value in the CSV matches what you see in the photo.	/
UNSURE (number key 2)	You cannot reach a confident conclusion. The record will be excluded from the clean dataset.	Image is blurry, too dark, gauge is cut off, or there is a large and unexplained difference between the tipping bucket and cup readings.
WRONG (number key 3)	The image is clear and readable, but the value recorded in the CSV is different from what the photo shows.	You can clearly read the gauge in the photo, but the number does not match the CSV. Enter the correct value in the box that appears. 

Confirming your decision

Once you have pressed a decision button (and entered a corrected value if applicable), click the **Confirm** → button at the bottom of the left panel. You can also press **Enter** on your keyboard instead of clicking. After confirming, the tool automatically moves to the next record.

Note: Your decision is not saved until you click Confirm (or press Enter). Pressing a colour button alone does not save anything.

5.4 The Notes Field

Below the decision buttons is an optional Notes field. You can type any observations here — for example, why you marked something as UNSURE, or a specific detail you noticed in the photograph. Notes are saved in the training_notes.json output file and are useful for future

reference, or for training for AI tools. The Notes field is completely optional. You can leave it blank for any or all records.

5.5 Navigating Between Records

Use the **← Prev** and **Next →** buttons at the bottom of the left panel to move between records. You can go back to review or change a previous decision at any time. The **Prev** button is not available on the first record, and **Next** is not available on the last.

Records are ordered by date, then by station name. The tool always starts from the first record when you open a session.

5.6 The Image Panel

The right side of the screen displays the photograph(s) for the current record. If there are two images, they are shown side by side with a label above each one (IMAGE 1 and IMAGE 2) and the filename displayed at the top of the panel. Use the zoom buttons or double click on an image to zoom.

If an image cannot be found in the selected folder, a warning will appear in place of the photograph: “⚠ Not found” followed by the filename. In this case, check that the filename in the CSV matches the file in the folder exactly, and that the correct folder was selected. A record with no image entry, but with a tipping bucket reading will still be shown.

For split readings, all images from all sub-readings are displayed together in the image panel.

6. Ending a Session and Exporting Results

Click the **End & Export** button in the top-right corner at any time. You do not need to have reviewed all records before exporting.

A summary window will appear showing:

- The total number of records in the dataset
- How many you have reviewed so far
- The count for each decision type (MATCH, UNSURE, WRONG)
- How many records remain unreviewed
-

Click **Export All Files** to download the three output files. If you want to continue reviewing instead, click **Cancel** to close the window and return to the review screen.

Note: Unreviewed records are excluded from all output files. If you stop halfway through, you can continue from where you left off — but you will need to reload the files again from the setup screen, and your previous decisions will not be remembered. Do not close the tab without exporting – you will lose all your work!

6.1 Output Files

Three files are downloaded automatically when you click Export All Files:

File	Contents	Description
<code>cleaned_data_YYYY-MM-DD_HH-MM.csv</code>	Clean readings	Contains only the verified records labelled MATCH or WRONG (with your corrected values). Use this file for analysis. The timestamp in the filename shows when it was exported.
<code>summary_table_YYYY-MM-DD_HH-MM.csv</code>	QC summary	A table showing how many records were reviewed per station, and how many received each label.
<code>training_notes_YYYY-MM-DD_HH-MM.json</code>	Label archive	A detailed record of every decision you made, including corrected values, notes, and timestamps. This file can be used to train automatic image-checking tools in the future.

Tip: Your browser may ask where to save the files, or it may save them automatically to your Downloads folder, depending on your browser settings.

7. Troubleshooting

The Start Review button does not activate

You must load at least the Gauge CSV and the Image Folder. Check that both show a green tick (✓). The Bucket CSV is optional.

Images show “⚠ Not found”

The filename in the gauge CSV does not match the actual file in the folder. Check for: wrong capitalisation, missing .jpg extension, extra spaces, or the wrong folder selected. Re-selecting the correct folder requires reloading the page.

The tool shows no records after loading

The gauge CSV may have no valid rows, or all rows may have been matched to bucket data only. Check that the gauge CSV has the correct column headers and at least one row of data with an image filename.

The date column shows numbers instead of dates

Excel sometimes converts dates automatically to a number format. Use the formula `=TEXT(A2, "YYYY-MM-DD")` in a new column to convert it to the correct text format, then copy and paste the result as values before saving as CSV.

The Confirm button does not respond

Make sure you have clicked one of the three decision buttons (MATCH, UNSURE, WRONG) first. If you selected WRONG, make sure you have entered a value in the corrected value box before clicking Confirm.

The tool opened in the wrong browser

Close the tab. Right-click the `precip_checker.html` file, choose Open with, and select Google Chrome or Microsoft Edge.

Annex: Preparing input file for tipping bucket readings

This document explains how to convert raw tipping bucket logger data (exported from the logger software as an Excel file) into the `bucket_input.csv` file required by the PrecipQC tool.

Two methods are provided. Method A uses a Python script that processes the data automatically. Method B achieves the same result using Microsoft Excel formulas only, with no programming required. Both produce identical output.

Method A: Python Script

This method runs a script that processes the Excel file automatically. It requires a one-time installation of Python and Visual Studio Code, which takes around 15–20 minutes the first time.

A1. Install Python

1. Open your web browser and go to: <https://www.python.org/downloads/>
2. Click the large yellow **Download Python** button. This downloads the installer for the latest version.
3. Run the downloaded installer. On the first screen, tick the box that says **Add Python to PATH** before clicking anything else. This box is near the bottom of the installer window. If you miss it, uninstall and reinstall.
4. Click **Install Now** and wait for the installation to finish.
5. Click **Close** when done.

Tip: To check Python installed correctly: press the Windows key, type `cmd`, press Enter. In the black window that opens, type `python --version` and press Enter. You should see something like Python 3.12.0. If you see an error, restart your computer and try again.

A2. Install Visual Studio Code

Visual Studio Code (VS Code) is a free text editor that makes it easy to run Python scripts.

6. Go to: <https://code.visualstudio.com/>
7. Click **Download for Windows** and run the installer.
8. Accept the default options throughout. On the “Select Additional Tasks” screen, tick **Add to PATH** if it is not already ticked.
9. Click **Install**, then **Finish**.
10. Open VS Code from the Start menu or desktop shortcut.
11. Install the Python extension inside VS Code:
 - a. Click the Extensions icon on the left sidebar (it looks like four squares).
 - b. In the search box, type `Python`.
 - c. Click the extension published by **Microsoft** and click **Install**.

A3. Install the Required Python Libraries

The script uses several Python libraries that are not included by default. You install them once using the Terminal inside VS Code.

12. In VS Code, open the Terminal by clicking **Terminal** in the top menu bar, then **New Terminal**. A panel opens at the bottom of the screen.
13. Type the following command and press Enter. Wait for it to finish (it may take a minute or two):

```
pip install pandas openpyxl scipy numpy matplotlib xarray
```

14. You should see a series of messages ending in *Successfully installed* If you see an error saying `pip is not recognised`, try replacing `pip` with `pip3` and running again.

Note: You only need to do this installation once. The next time you use the script, skip straight to Part A4.

A4. Prepare Your Files

15. Create a new folder on your computer (for example on your Desktop) and name it something like *bucket_processing*.
16. Place your tipping bucket Excel file inside this folder. Rename it to exactly `TippingBucket.xlsx` (the script looks for this exact filename).
17. Save the Python script as a new file named `process_bucket.py` inside the same folder. To do this:
 - d. In VS Code, click **File** → **New File**.
 - e. Paste the full script code into the new file.
 - f. Click **File** → **Save As**, navigate to your `bucket_processing` folder, and save the file as `process_bucket.py`.

Your folder should now contain exactly:

```
bucket_processing/  
  TippingBucket.xlsx  
  process_bucket.py
```

A5. Open the Folder in VS Code and Run the Script

18. In VS Code, click **File** → **Open Folder** and select your `bucket_processing` folder. Click **Select Folder**.
19. In the Explorer panel on the left, click `BucketPrep.py` to open it.
20. Open the Terminal (**Terminal** → **New Terminal**).
21. In the Terminal, type the following and press Enter:

```
python BucketPrep.py
```

22. Wait for the script to finish. When it is done, the terminal prompt will reappear with no error messages. A new file called `bucket_input.csv` will appear in your folder.
23. Open `bucket_input.csv` in Excel to check it looks correct. It should have four columns: `location`, `date`, `tips`, `bucket_mm`. Each row is one station on one day.

Tip: If you see an error message in the Terminal, read it carefully. The most common errors are: the Excel file is not named exactly TippingBucket.xlsx, or the file is open in Excel while the script is running. Close Excel and try again.

A6. Understanding the Output Columns

Term / Command	What it means
<code>location</code>	The station name, taken directly from the sheet name in your Excel file.
<code>date</code>	The calendar date of the reading, in YYYY-MM-DD format. Timestamps after 7:00 PM are assigned to the following day.
<code>tips</code>	The number of valid tips (events with value = 1) recorded at that station on that date.
<code>bucket_mm</code>	Total rainfall in millimetres: number of tips × 0.2 mm per tip.

Note: If your tipping bucket records a different volume per tip (e.g. 0.5 mm instead of 0.2 mm), open the script in VS Code, find the line that reads `tips_daily['bucket_mm'] = tips_daily['tips'] * 0.2` and change 0.2 to your correct value. Save the file and run the script again.

Method B: Microsoft Excel Only

This method does not require Python or any programming. It uses Excel formulae and built-in tools to produce the same output. It takes longer to set up, is more prone to manual errors but requires no software installation.

Important: This method works best if your logger data has the same structure across all station sheets. If the column names or layouts are very inconsistent between sheets, the Python script (Method A) will handle this more reliably.

B1. Understand Your Source File

Your Excel file (`TippingBucket.xlsx`) has one sheet per station. On each sheet, there is a column for the timestamp (when the tip occurred) and a column for the event value (1 = real tip, 0 = other). The column names may vary slightly between sheets (e.g. “Time/Date” or “Time”; “Event” or “Event Type”).

Before starting, open the file and note down:

- The exact name of the timestamp column on each sheet (e.g. “Time/Date”)
- The exact name of the event column (e.g. “Event”)
- Which row the data starts on (usually row 2, after a header row)

B2. Prepare Each Station Sheet

Work through every station sheet one at a time. For each sheet, you will add three columns: the station name, an adjusted date, and a valid tip flag.

Add the station name column

1. Find an empty column to the right of your data. Label it location in the header row.
2. In the first data row, type the station name exactly as you want it to appear in the output (e.g. Tisunayili).
3. Copy that cell down to all data rows. The quickest way: click the cell, copy it (Ctrl+C), then select all the rows below it in that column and paste (Ctrl+V).

Add the adjusted date column

4. In the next empty column, add the header `adjusted_date`.
5. In the first data row, enter the following formula, replacing B2 with the cell containing the timestamp for that row:

```
=INT(B2 + 17/24)
```

This adds 17 hours to the timestamp, which shifts the 7 AM morning cutoff to midnight. Tips recorded before 7 AM are thereby grouped with the previous calendar day, matching the observer's morning reading routine. `INT()` then strips the time portion, leaving only the date.

6. Format the `adjusted_date` column as a date: select the column, right-click → **Format Cells** → **Custom** → type YYYY-MM-DD in the box → click OK. The cells should now show dates like 2025-06-02. If they still show numbers (e.g. 45695), the format has not been applied correctly — repeat this step.
7. Copy the formula down to all data rows.

Add the valid tip column

8. In the next empty column, add the header `valid_tip`.
9. In the first data row, enter the following formula, replacing C2 with the cell containing the event value for that row:

```
=IF(C2=1, 1, 0)
```

10. Copy the formula down to all data rows.

Repeat for all stations before moving to the next step. Every sheet should now have three new columns: `location`, `adjusted_date`, and `valid_tip`.

B3. Combine All Sheets Into One

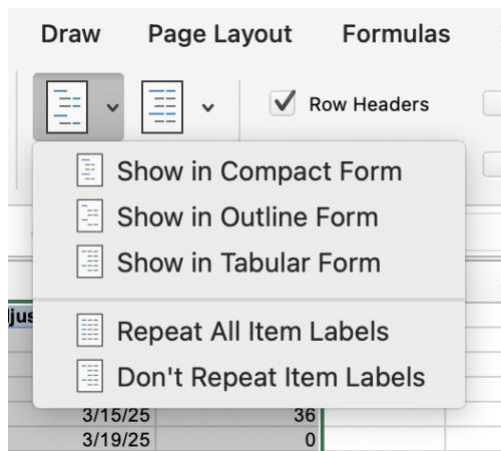
You will combine all the data into a single sheet first, then build one Pivot Table covering all stations at once.

1. Add a new sheet to the workbook and name it `all_data`.
2. Go to the first station sheet. Select all the data including the header row (click the top-left cell, then press Ctrl+Shift+End to extend the selection to the last used cell). Copy it (Ctrl+C).
3. Go to the `all_data` sheet and paste into cell A1.

4. Go to the second station sheet. Select all the data **excluding the header row** (start from row 2). Copy and paste it into the `all_data` sheet, immediately below the last row of data already there.
5. Repeat for all remaining stations. The `all_data` sheet should end up with one header row at the top and all tips from all stations below it, with the location column identifying which station each row belongs to.

B4. Build One Pivot Table for All Stations

1. Click anywhere inside the data on the `all_data` sheet.
2. Go to **Insert** → **PivotTable or PivotChart** → select relevant columns → choose **New Worksheet** → click OK.
3. In the PivotTable Fields panel on the right:
 - Drag `location` to the **Rows** area.
 - Drag `adjusted_date` to the **Rows** area, below location.
 - Drag `valid_tip` to the **Values** area. Make sure it says **Sum of valid_tip**. If it says **Count**, click the field → **Value Field Settings** → choose **Sum**.
4. Remove station-level subtotals:
 - Right-click any station name in the pivot table, → select **Field Settings**.
 - In the dialog, go to the **Subtotals & Filters** tab, → under Subtotals, select **None**, → click **OK**.
5. Enable tabular layout and repeat station names:
 - With the pivot table selected, go to **the PivotTable Analyze** tab (or Analyze or Design tab) in the ribbon.
 - Click the **Report Layout** dropdown button, → choose **Show in Tabular Form**.



- Click **Report Layout** again, → choose **Repeat All Item Labels**.
6. Verify dates are ungrouped:
 - If you see months/years instead of individual dates, right-click any date in the pivot table, → select **Ungroup**.
 7. Click **Refresh** (in the PivotTable Analyze tab) to ensure all data and settings are up to date.

	A	B	C
1	location	Adjusted_date	Sum of Valid_tip
2	Galinkpegu	2/24/25	1
3	Galinkpegu	3/6/25	2
4	Galinkpegu	3/13/25	8
5	Galinkpegu	3/15/25	36
6	Galinkpegu	3/19/25	0
7	Galinkpegu	4/10/25	3
8	Galinkpegu	4/11/25	47
9	Galinkpegu	4/17/25	18
10	Galinkpegu	4/19/25	1
11	Galinkpegu	4/23/25	2
12	Galinkpegu	4/27/25	1
13	Galinkpegu	4/28/25	8
14	Galinkpegu	4/29/25	213
15	Tingoli	2/7/25	23
16	Tingoli	3/19/25	2
17	Tingoli	4/11/25	63
18	Tingoli	4/18/25	11
19	Tingoli	4/23/25	2
20	Tingoli	4/27/25	4
21	Tingoli	4/29/25	512
22	Tingoli	4/30/25	1

Figure 3. End result after step B4

B5. Build the Output Table

1. Copy the entire Pivot Table result — the location names, dates, and tip sums. Do not include the grand total row if one appears.
2. Paste as values only into a new sheet named `combined`: right-click the target cell → **Paste Special** → **Values** → OK. This removes the Pivot Table formatting and leaves plain data.
3. Rename the columns to match the required output format. Your four columns should be:

```
location | date | tips | bucket_mm
```

4. The `bucket_mm` column does not come from the Pivot Table — you need to add it. In the first data row of the `bucket_mm` column, enter:

```
=C2*0.2
```

Replace C2 with the cell containing the tip count for that row. Each tip equals 0.2 mm of rainfall, so this formula converts the tip count to rainfall in mm. Double click the green square at the bottom right of the cell to autofill the formula to all rows below.

5. Check that the date column is formatted as YYYY-MM-DD. If it shows numbers (e.g. 45695), select the column → right-click → **Format Cells** → **Custom** → type YYYY-MM-DD → OK.
6. Sort the data by date, then by location: go to **Data** → **Sort**. Add two levels — first by date (A to Z), then by location (A to Z).

B6. Save as CSV

1. Click the `combined` sheet tab to make sure it is the active sheet.
2. Ctrl + A to copy all cells of the file. Open a new Excel workbook, and paste the cells you copied.
3. In this new workbook, go to **File** → **Save As**.
4. Name the file `bucket_input.csv`.
5. In the **Save as type** dropdown, select **CSV (Comma delimited) (*.csv)**.
6. Click **Save**. Excel might warn you that CSV cannot save multiple sheets — click **OK** or **Keep Current Format**.
7. Open the saved file in Notepad (right-click → Open with → Notepad) to verify the first few lines look like:

```
location,date,tips,bucket_mm
Garizegu,2025-06-02,12,2.4
Tisunayili,2025-06-02,5,1.0
Sanga,2025-06-03,3,0.6
```

Important: If the date column appears as a number in the CSV (e.g. 45695), go back to the Excel file, re-format the date column as custom YYYY-MM-DD, then re-save as CSV. Excel sometimes loses date formatting when saving to CSV.

Tip: To quickly check whether all dates are in the correct format, open the CSV in Excel and look for any cells that appear right-aligned (numbers) in the date column. Dates stored as text will be left-aligned.